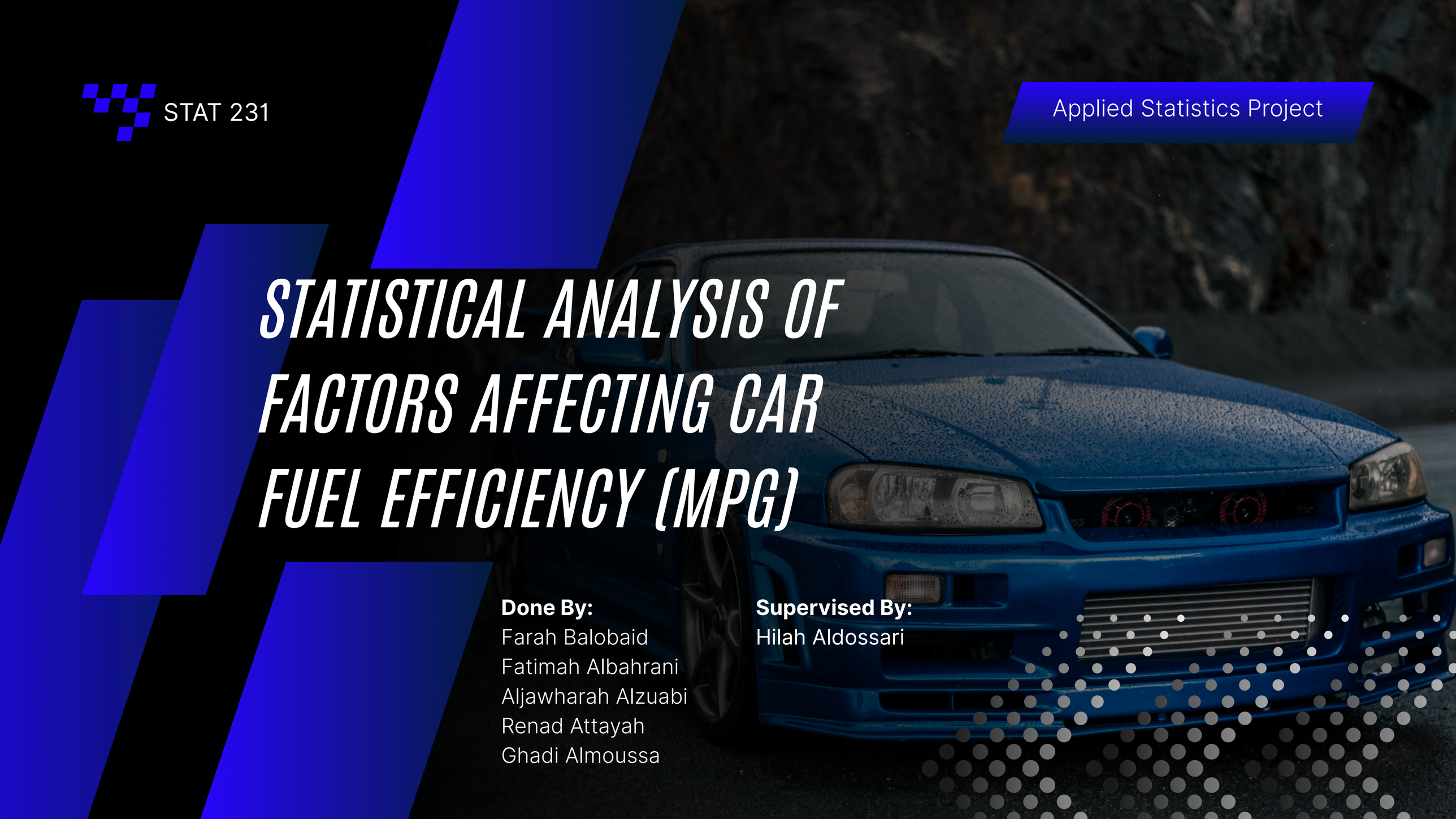
STAT 231

Applied Statistics Project

STATISTICAL ANALYSIS OF FACTORS AFFECTING CAR FUEL EFFICIENCY (MPG)



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STAT 231

BRIEF DESCRIPTION

Collect statistics about a random sample of 28 cars to understand how their physical features (like engine power, weight, and cylinder count) affect their fuel efficiency (MPG).

The project aims to analyze the current data, find trends, and identify the most important factors that lead to better or worse gas mileage.



STAT 231

CONTENT

- BRIEF DESCRIPTION
- THE DATA SET
- RESEARCH QUESTIONS
- DATA DESCRIPTION



STAT 231

THE DATA SET

mpg	cylinders	horsepower	weight	acceleration	model_year	origin
27.4	4	106	2954	16.9	76	Europe
32	8	74	3453	20.5	82	Japan
28.9	8	140	3913	17.5	78	USA
27.3	4	175	4004	14.3	78	Europe
23.9	8	142	2320	21.7	73	Europe
30.1	4	140	3156	9.6	72	Europe
24.3	4	179	3876	18.7	73	Europe
37	4	134	3022	13.8	80	Europe
39	6	116	4000	17.5	82	Japan
22.7	6	122	4011	11.5	76	Japan
34.2	8	139	1999	12.4	73	Japan
26.8	4	107	2467	21.5	76	USA
27.9	4	144	1822	16.2	75	Europe
37.9	6	105	3822	17.2	77	Japan
14	4	156	3353	15.5	81	Europe
14.4	6	174	3819	11.2	70	Europe
12.6	8	143	2435	13.5	81	Europe
35.3	8	71	2755	16.7	78	Europe
33.8	4	139	3127	14.6	74	USA
36.4	6	144	2196	14.6	80	Japan
39.4	6	177	2688	18	76	Japan
34.4	6	167	2151	12	81	USA
24.9	6	126	3167	13.4	75	Europe
33.9	8	76	3489	18.5	82	Europe
15.3	8	120	2492	20.3	78	USA
29.9	8	72	2840	9.4	72	USA
16	4	119	3615	9.3	81	USA
38.5	8	178	2751	13	73	Japan

STAT 231

ANALYSIS QUESTIONS:







Q1

What are the descriptive statistics, frequency distributions, and graphical summaries for the variables in the dataset?

Q2

is there a significant linear relationships between a car's weight and its fuel efficiency (MPG)

Q3

Is the average Fuel Efficiency (MPG) for the sample of cars significantly different from the industry benchmark of 28 MPG?

STAT 231

ANALYSIS QUESTIONS:



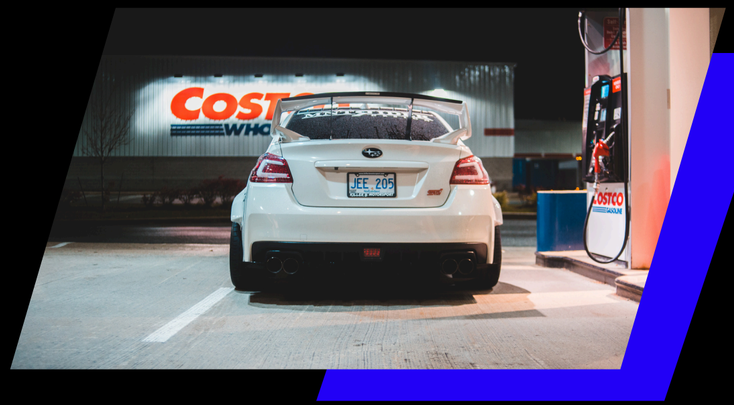
Q4

Does the average fuel efficiency (MPG) differ significantly among cars with 4,6, and 8 cylinders?



Q5

Is there a relationship between car weight (weight) and fuel efficiency (mpg)?



Q6

Do cars with horsepower above 140 HP have significantly different MPG than cars with horsepower 140 HP or below?

STAT 231

ANALYSIS QUESTIONS:



Q7

Which car model year has the highest avrerage fuel efficiency (MPG)?



Q8

Does the model year of the car significantly predict the MPG of the car?

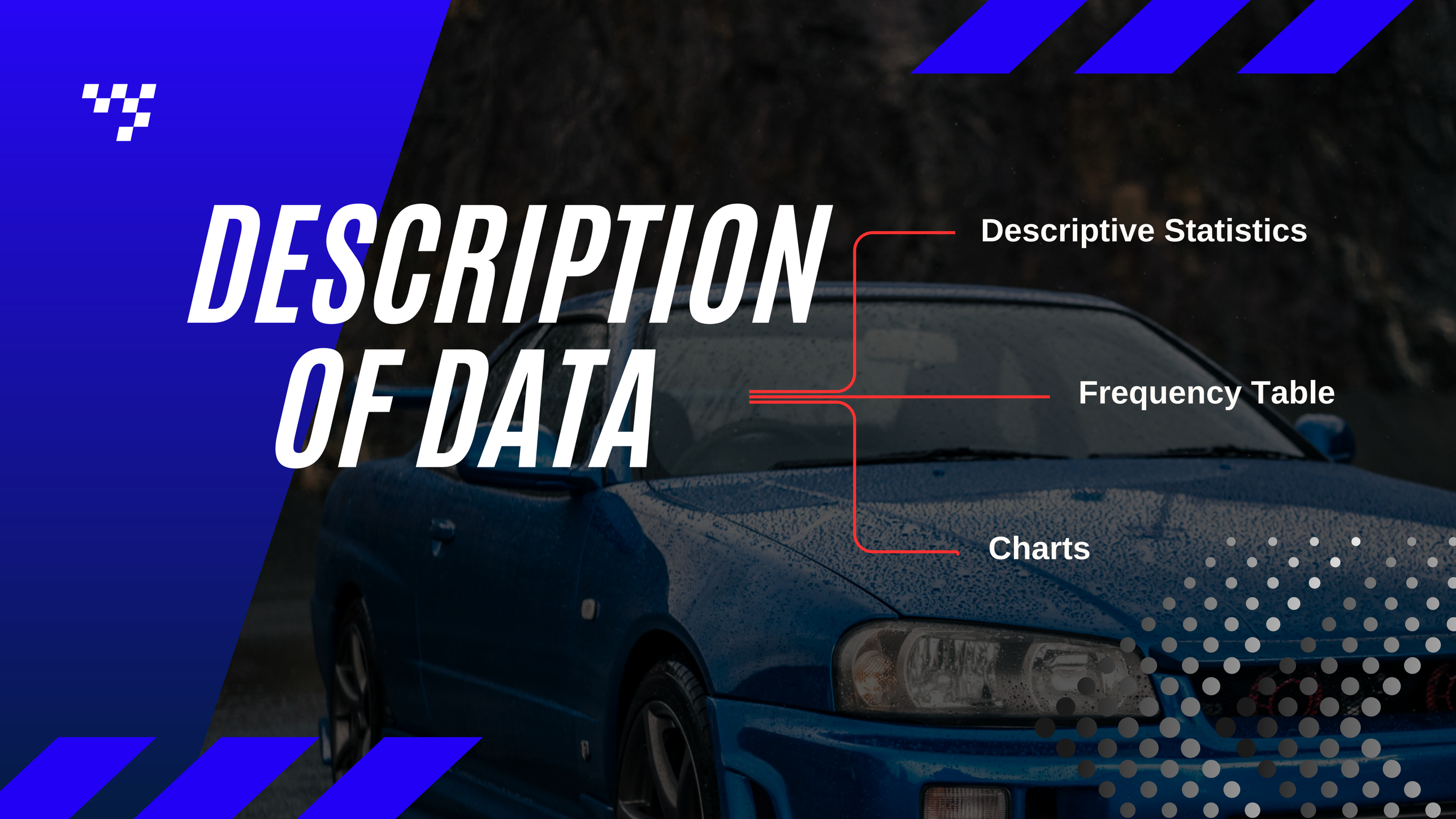


Q9

Which factor has the most influence on MPG: weight, horsepower, or cylinders?

DESCRIPTION OF DATA

- Descriptive Statistics
- Frequency Table
- Charts



Q1

DESCRIPTIVE STATISTICS

Statistics

		mpg	cylinders	horsepower	weight	acceleration	model_year	origin
N	Valid	28	28	28	28	28	28	28
	Missing	0	0	0	0	0	0	0
Mean		28.507	6.00	131.61	3060.61	15.332	76.89	
Median		29.400	6.00	139.00	3074.50	15.050	76.50	
Mode		12.6[a]	4[a]	139[a]	1822[a]	14.6[a]	73[a]	
Std. Deviation		8.2180	1.721	32.804	672.707	3.5972	3.573	
Variance		67.536	2.963	1076.099	452534.988	12.940	12.766	
Range		26.8	4	108	2189	12.4	12	

a. Multiple modes exist. The smallest value is shown

- **The following tables show the distribution of values for our seven variables: MPG, Cylinders, Horsepower, Weight, Acceleration, Model Year, and Origin.**
- **All variables have 28 valid entries with no missing data. Each table includes Frequency, Percent, Valid Percent, and Cumulative Percent.**

Q1

FREQUENCY TABLE

- The following tables show the distribution of values for our seven variables: MPG, Cylinders, Horsepower, Weight, Acceleration, Model Year, and Origin.

- All variables have 28 valid entries with no missing data. Each table includes Frequency, Percent, Valid Percent, and Cumulative Percent.

mpg

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	12.6	1	3.6	3.6	3.6
	14.0	1	3.6	3.6	7.1
	14.4	1	3.6	3.6	10.7
	15.3	1	3.6	3.6	14.3
	16.0	1	3.6	3.6	17.9
	22.7	1	3.6	3.6	21.4
	23.9	1	3.6	3.6	25.0
	24.3	1	3.6	3.6	28.6
	24.9	1	3.6	3.6	32.1
	26.8	1	3.6	3.6	35.7
	27.3	1	3.6	3.6	39.3
	27.4	1	3.6	3.6	42.9
	27.9	1	3.6	3.6	46.4
	28.9	1	3.6	3.6	50.0
	29.9	1	3.6	3.6	53.6
	30.1	1	3.6	3.6	57.1
	32.0	1	3.6	3.6	60.7
	33.8	1	3.6	3.6	64.3
	33.9	1	3.6	3.6	67.9
	34.2	1	3.6	3.6	71.4
	34.4	1	3.6	3.6	75.0
	35.3	1	3.6	3.6	78.6
	36.4	1	3.6	3.6	82.1
	37.0	1	3.6	3.6	85.7
	37.9	1	3.6	3.6	89.3
	38.5	1	3.6	3.6	92.9
	39.0	1	3.6	3.6	96.4
	39.4	1	3.6	3.6	100.0
	Total	28	100.0	100.0	

horsepower

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	71	1	3.6	3.6	3.6
	72	1	3.6	3.6	7.1
	74	1	3.6	3.6	10.7
	76	1	3.6	3.6	14.3
	105	1	3.6	3.6	17.9
	106	1	3.6	3.6	21.4
	107	1	3.6	3.6	25.0
	116	1	3.6	3.6	28.6
	119	1	3.6	3.6	32.1
	120	1	3.6	3.6	35.7
	122	1	3.6	3.6	39.3
	126	1	3.6	3.6	42.9
	134	1	3.6	3.6	46.4
	139	2	7.1	7.1	53.6
	140	2	7.1	7.1	60.7
	142	1	3.6	3.6	64.3
	143	1	3.6	3.6	67.9
	144	2	7.1	7.1	75.0
	156	1	3.6	3.6	78.6
	167	1	3.6	3.6	82.1
	174	1	3.6	3.6	85.7
	175	1	3.6	3.6	89.3
	177	1	3.6	3.6	92.9
	178	1	3.6	3.6	96.4
	179	1	3.6	3.6	100.0
	Total	28	100.0	100.0	

Q1

FREQUENCY TABLE

- **weight: 28 distinct weights, all with equal frequency. Shows a wide range of vehicle weights.**
- **acceleration: Mostly unique values, with one repeated (14.6). Distribution covers low to high acceleration.**

weight

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1822	1	3.6	3.6	3.6
	1999	1	3.6	3.6	7.1
	2151	1	3.6	3.6	10.7
	2196	1	3.6	3.6	14.3
	2320	1	3.6	3.6	17.9
	2435	1	3.6	3.6	21.4
	2467	1	3.6	3.6	25.0
	2492	1	3.6	3.6	28.6
	2688	1	3.6	3.6	32.1
	2751	1	3.6	3.6	35.7
	2755	1	3.6	3.6	39.3
	2840	1	3.6	3.6	42.9
	2954	1	3.6	3.6	46.4
	3022	1	3.6	3.6	50.0
	3127	1	3.6	3.6	53.6
	3156	1	3.6	3.6	57.1
	3167	1	3.6	3.6	60.7
	3353	1	3.6	3.6	64.3
	3453	1	3.6	3.6	67.9
	3489	1	3.6	3.6	71.4
	3615	1	3.6	3.6	75.0
	3819	1	3.6	3.6	78.6
	3822	1	3.6	3.6	82.1
	3876	1	3.6	3.6	85.7
	3913	1	3.6	3.6	89.3
	4000	1	3.6	3.6	92.9
	4004	1	3.6	3.6	96.4
	4011	1	3.6	3.6	100.0
	Total	28	100.0	100.0	

acceleration

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	9.3	1	3.6	3.6	3.6
	9.4	1	3.6	3.6	7.1
	9.6	1	3.6	3.6	10.7
	11.2	1	3.6	3.6	14.3
	11.5	1	3.6	3.6	17.9
	12.0	1	3.6	3.6	21.4
	12.4	1	3.6	3.6	25.0
	13.0	1	3.6	3.6	28.6
	13.4	1	3.6	3.6	32.1
	13.5	1	3.6	3.6	35.7
	13.8	1	3.6	3.6	39.3
	14.3	1	3.6	3.6	42.9
	14.6	2	7.1	7.1	50.0
	15.5	1	3.6	3.6	53.6
	16.2	1	3.6	3.6	57.1
	16.7	1	3.6	3.6	60.7
	16.9	1	3.6	3.6	64.3
	17.2	1	3.6	3.6	67.9
	17.5	2	7.1	7.1	75.0
	18.0	1	3.6	3.6	78.6
	18.5	1	3.6	3.6	82.1
	18.7	1	3.6	3.6	85.7
	20.3	1	3.6	3.6	89.3
	20.5	1	3.6	3.6	92.9
	21.5	1	3.6	3.6	96.4
	21.7	1	3.6	3.6	100.0
	Total	28	100.0	100.0	

Q1

FREQUENCY TABLE

cylinders

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	4	10	35.7	35.7	35.7
	6	8	28.6	28.6	64.3
	8	10	35.7	35.7	100.0
	Total	28	100.0	100.0	

origin

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Europe	13	46.4	46.4	46.4
	Japan	8	28.6	28.6	75.0
	USA	7	25.0	25.0	100.0
	Total	28	100.0	100.0	

model_year

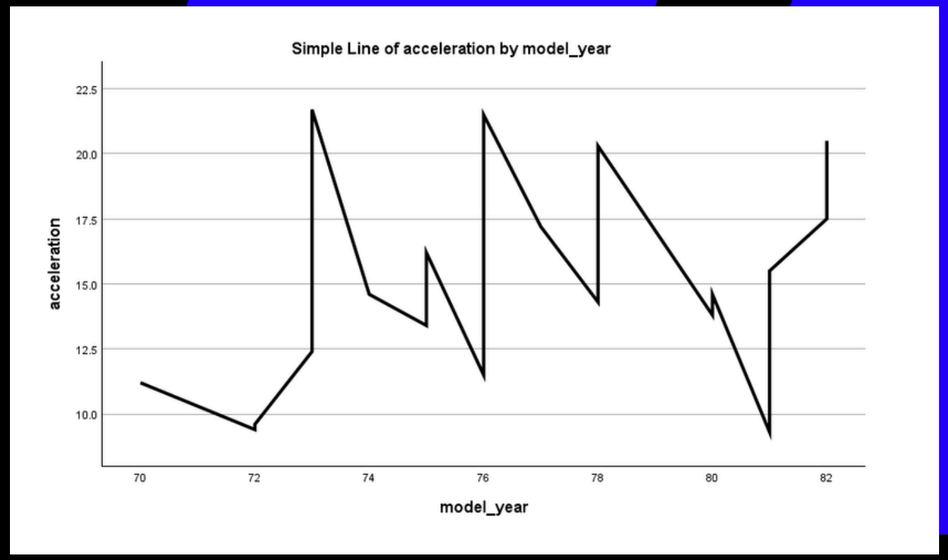
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	70	1	3.6	3.6	3.6
	72	2	7.1	7.1	10.7
	73	4	14.3	14.3	25.0
	74	1	3.6	3.6	28.6
	75	2	7.1	7.1	35.7
	76	4	14.3	14.3	50.0
	77	1	3.6	3.6	53.6
	78	4	14.3	14.3	67.9
	80	2	7.1	7.1	75.0
	81	4	14.3	14.3	89.3
	82	3	10.7	10.7	100.0
	Total	28	100.0	100.0	

- **cylinders: Three categories (4, 6, 8) with balanced counts.**
- **origin: Europe leads (46.4%), followed by Japan and USA.**
- **model_year: Spread across 11 years, with 1973, 1976, 1978, and 1981 most frequent.**

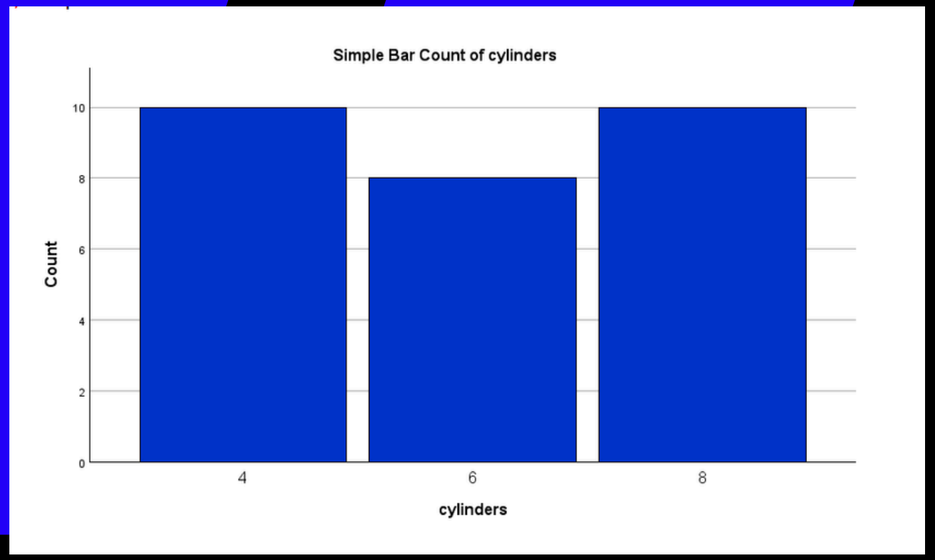
Q1

CHARTS

LINE CHART (ACCELERATION)



BAR CHART (CYLINDERS)

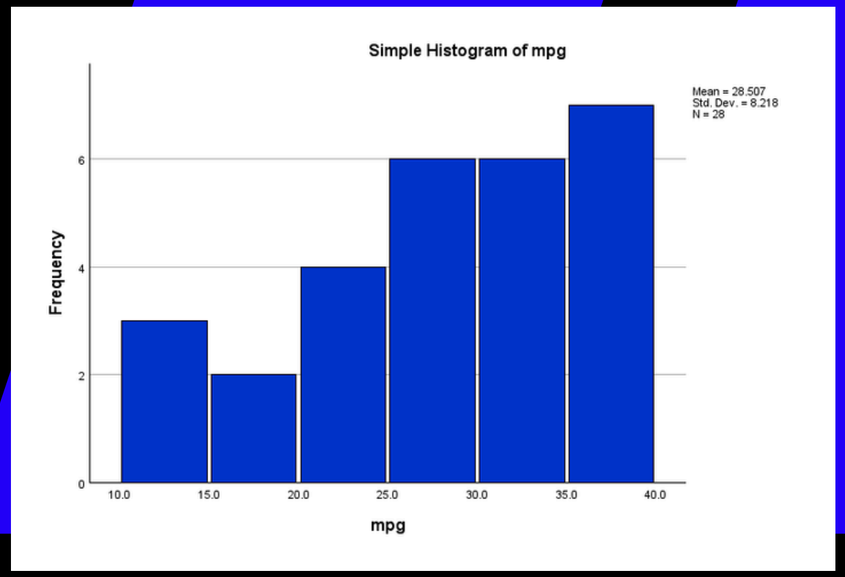


- Line chart: Acceleration trends across model years.
- Bar chart: Count of vehicles by cylinder number — 4 and 8 cylinders are most common.

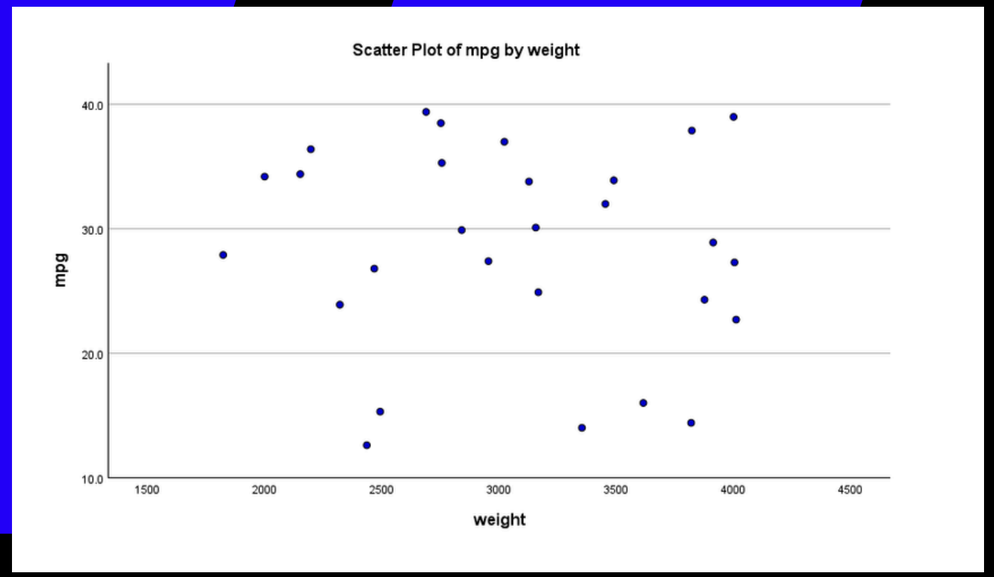
Q1 + Q2

CHARTS

HISTOGRAM (MPG)



SCATTERPLOT (WEIGHT)

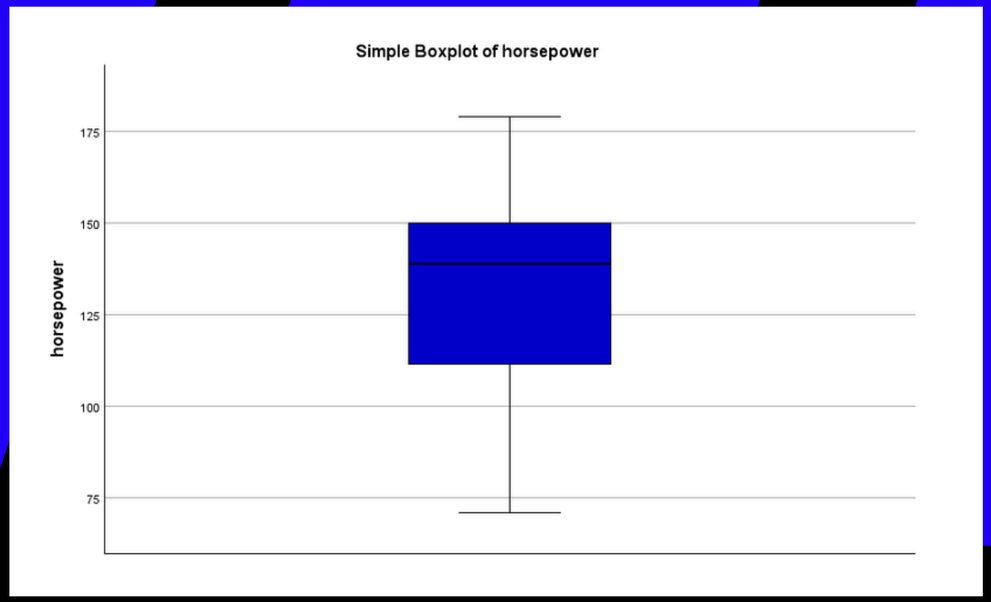


- Histogram (mpg): Most vehicles fall in the 30–40 mpg range. Mean = 25.15.
- Scatterplot (weight vs mpg): Shows negative correlation — heavier cars tend to have lower mpg.

Q1

CHARTS

BOXPLOT (HORSEPOWER)



- **Boxplot: Displays horsepower distribution. Useful for spotting median, spread, and outliers.**

Q3

IS THE AVERAGE FUEL EFFICIENCY (MPG) FOR THE SAMPLE OF CARS SIGNIFICANTLY DIFFERENT FROM THE INDUSTRY BENCHMARK OF 28 MPG?

1 TEST OF NORMALITY

Tests of Normality

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
mpg	.133	28	.200*	.922	28	.038

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Test Name	Test of Normality	
H_0 **H1**	Population is Normal Population is Not Normal	
Output	Kolmogorov 0.2	Shapiro 0.038
Decision	Since P=0.2 > 0.1 , then we **accept** Ho . So the population is **normal.**	

Q3

2 ONE-SAMPLE T-TEST

One-Sample Test

Test Value = 28

	t	df	Significance One-Sided p	Significance Two-Sided p	Mean Difference	95% Confidence Interval of the Difference Lower	95% Confidence Interval of the Difference Upper
mpg	.327	27	.373	.747	.5071	-2.679	3.694

Assumptions	• Population is normal. • Population variance is unknown. • Sample is simple random sample • Sample size < 30
H_0 **H1**	H_0: μ = 28 $H1$: $\mu \neq 28$
Test Name	One Sample T-Test
Output	Significance: One-Sided p = .373, Two-Sided p = .747
Decision	Since P=0.747 > 0.05 , then we **accept** Ho . and decide that average = 28 under α = 0.05

Q4

DOES THE AVERAGE FUEL EFFICIENCY (MPG) DIFFER SIGNIFICANTLY AMONG CARS WITH 4,6, AND 8 CYLINDERS?

1 TEST OF NORMALITY

INDEPENDENT VALUES

Tests of Normality

		Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	cylinders	Statistic	df	Sig.	Statistic	df	Sig.
mpg	4	.219	10	.190	.925	10	.402
	6	.262	8	.112	.850	8	.096
	8	.221	10	.183	.884	10	.145

a. Lilliefors Significance Correction

Test Name	Test of Normality	
H_0 **H1**	Population is Normal Population is Not Normal	
Output	kolmogorov 0.219 0.262 0.221	shapiro 0.925 0.850 0.884
Decision	Since P=0.2 > 0.1 , then we **accept** Ho . So the population is **normal.**	

Q4

2 ONE-WAY A NOVA

HOMO OF VARIANCES

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
mpg	Based on Mean	.871	2	25	.431
	Based on Median	.335	2	25	.718
	Based on Median and with adjusted df	.335	2	23.311	.719
	Based on trimmed mean	.764	2	25	.476

Assumptions	• Population is normal. • Population variance is unknown. • Sample is simple random sample • Sample size < 30
H_0 **H1**	H_0: $\mu = \mu$ $H1$: $\mu \neq \mu$
Test Name	One way a nova
Output	Based on Mean .871 2 25 .431 Based on Median .335 2 25 .718
Decision	Since P=0.871 > 0.05 , then we **accept** Ho which means there is no diff in variances

Q4

2 ONE-WAY A NOVA

FULL RESULTS

ANOVA

mpg

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	97.291	2	48.645	.705	.504
Within Groups	1726.168	25	69.047		
Total	1823.459	27			

Assumptions	• Population is normal. • Population variance is unknown. • Sample is simple random sample • Sample size < 30
H_0 **H1**	H_0: $\mu = \mu$ $H1$: $\mu \neq \mu$
Test Name	One way a nova
Output	F: .705, Sig.: .504
Decision	Since P=0.705 > 0.05 , then we **accept** Ho which means there is no diff in populatin means

Q5

IS THERE A RELATIONSHIP BETWEEN CAR WEIGHT AND FUEL EFFICIENCY (MPG)?, USE α = 0.05

1 TEST OF NORMALITY

Tests of Normality

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
mpg	.133	28	.200*	.922	28	.038
weight	.120	28	.200*	.947	28	.168

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Test Name	Test of Normality	
H_0 **H1**	Both Population are Normal At least one Population is Not Normal	
Output	Kolmogorov 0.2 0.2	Shapiro 0.038 0.168
Decision	Since P=0.2 > 0.1 , then we **accept** Ho . So both population are **normal.**	

Q5

2 PEARSON CORRELATIONS

Correlations

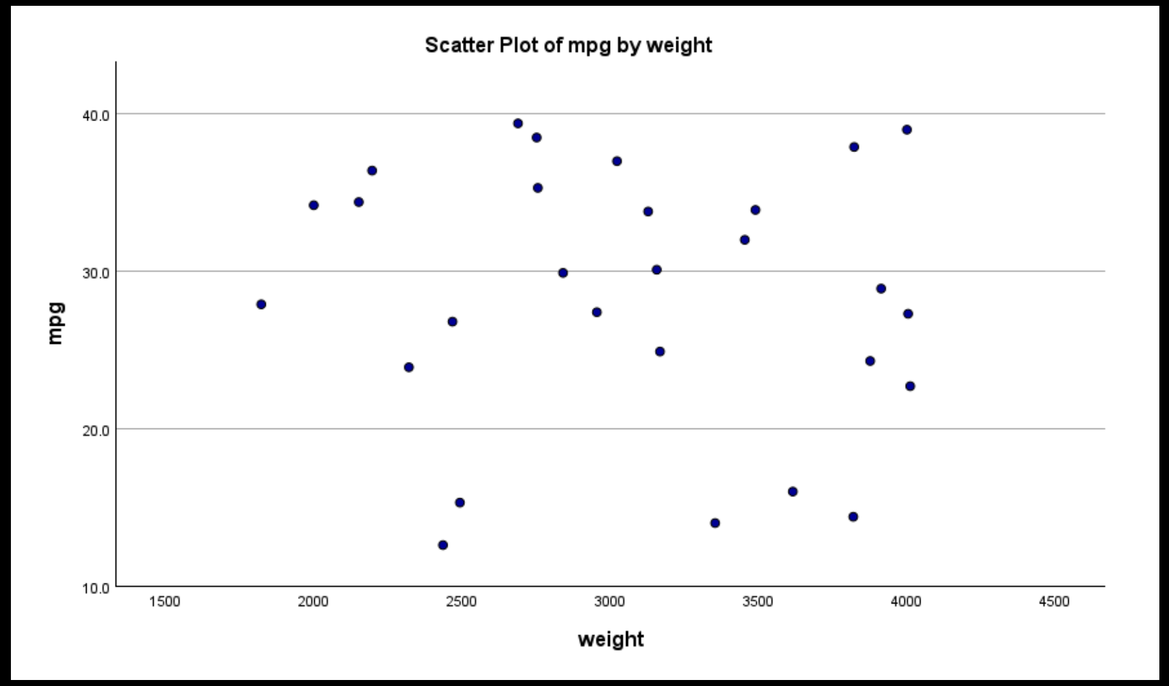
		mpg	weight
mpg	Pearson Correlation	1	-.108
	Sig. (2-tailed)		.584
	N	28	28
weight	Pearson Correlation	-.108	1
	Sig. (2-tailed)	.584	
	N	28	28

Statistical Model	Test of independence
H_0 **H1**	Two variables are independent $\rho=0$ Two variables are dependent $\rho\neq0$
Test Name	Pearson Correlation Coefficient
Output	p-value = 0.584 ρ = -0.108
Decision	Since P=0.584 > 0.05 , then we **accept** Ho . and decide the two variables are independent.

Since the Pearson Correlation = -0.108, then there is a weak negative relation between mpg & weight

Q5

3 GENERATE A SCATTER GRAPH



The scatterplot shows a negative relationship between weight and mpg. As the weight of the car increases, the mpg values tend to decrease. The points are not perfectly aligned, but the overall pattern clearly slopes downward, indicating this inverse relationship.

Q6

DO CARS WITH HORSEPOWER ABOVE 140 HP HAVE SIGNIFICANTLY DIFFERENT MPG THAN CARS WITH HORSEPOWER 140 HP OR BELOW?

1 TEST OF NORMALITY

Tests of Normality

		Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	=<140 vs >140	Statistic	df	Sig.	Statistic	df	Sig.
mpg	1.00	.133	17	.200*	.929	17	.211
	2.00	.165	11	.200*	.909	11	.238

Test Name	Test of Normality
H_0 **H1**	Both population are Normal At least one population is Not Normal
Output	Kolmogorov — Shapiro =<140 — 0.2 — 0.211 >140 — 0.2 — 0.238
Decision	Since P-value=0.2 > 0.1 , then we **accept** Ho . So the population are **normal.**

Q6

2 INDEPENDENT SAMPLES T-TEST

HOMO OF VARIANCES

Homogeneity of Variance Test

		Levene's Test for Equality of Variances	
		F	Sig.
mpg	Equal variances assumed	2.095	.160

Statiistical Model	About difference population mean
Assumptions	• Both populations are normal • Both variance are unknown and equal • Both sample are simple random sample • Two samples are independent • n1, n2 > 30
H_0 **H1**	H_0: $\mu = \mu$ H1: $\mu \neq \mu$
Test Name	Independent samples T-test
Output	Sig. .160
Decision	For variance test : since the P-value = 0.160>0.05, we accept null hypothesis , the we decide the variance for both populations are equal (accept).

Q6

2 INDEPENDENT SAMPLES T-TEST

FULL RESULTS

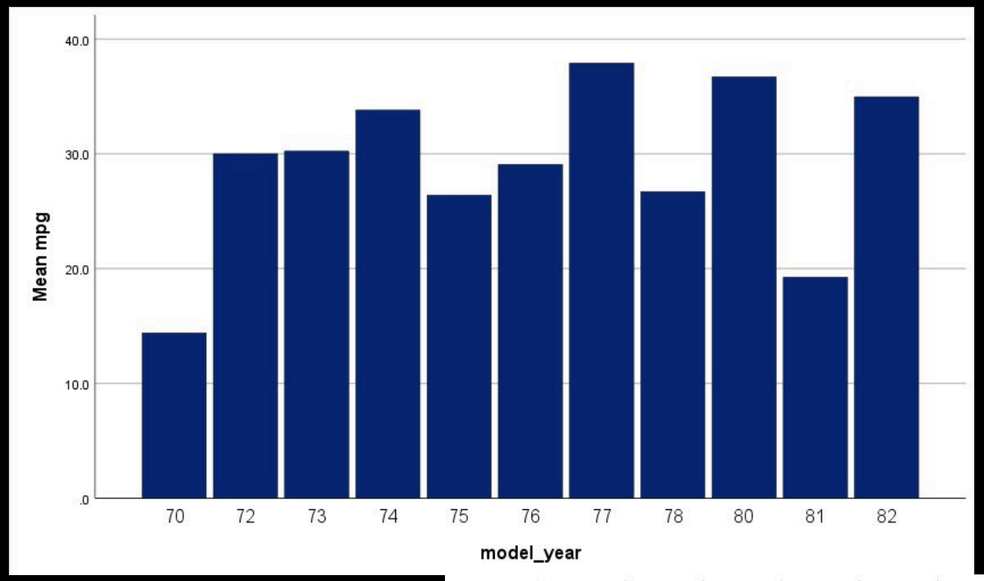
Independent Samples Test

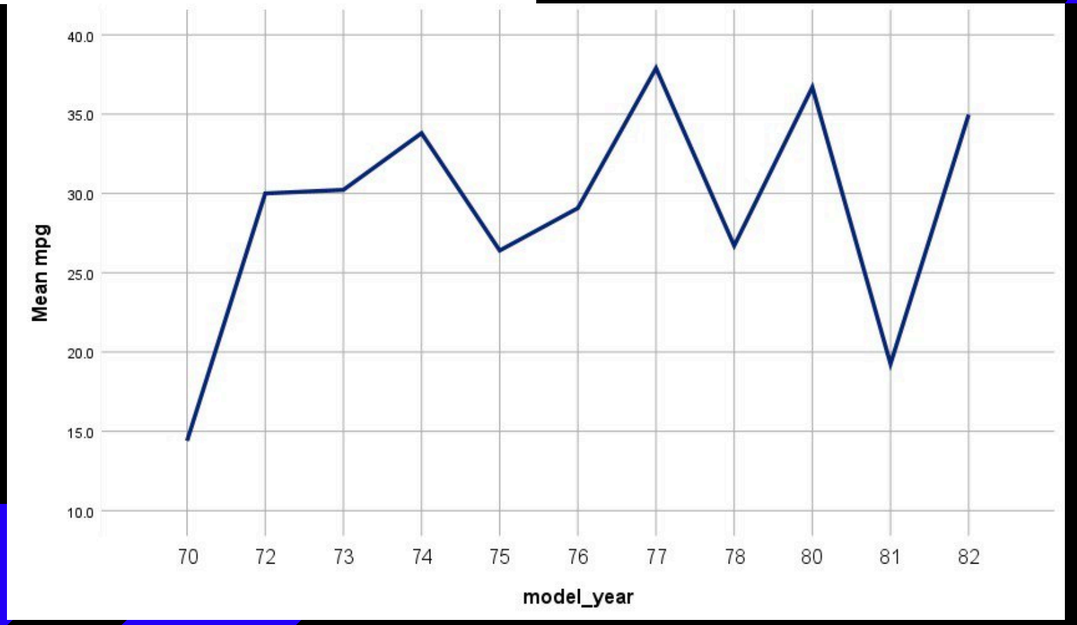
		t-test for Equality of Means							
				Significance				95% Confidence Interval of the Difference	
		t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
mpg	Equal variances assumed	.963	26	.172	.344	3.0663	3.1843	-3.4791	9.6117
	Equal variances not assumed	.893	16.407	.192	.385	3.0663	3.4326	-4.1959	10.3285

Statiistical Model	About difference population mean
Assumptions	• Both populations are normal • Both variance are unknown and equal • Both sample are simple random sample • Two samples are independent • n1, n2 > 30
H_0 **H1**	H_0: $\mu = \mu$ H1: $\mu \neq \mu$
Test Name	Independent samples T-test
Output	One-Sided p: .172, .192 Two-Sided p: .344, .385
Decision	For Independent samples test: since the P-value = 0.344>0.05, we accept the Ho , then we decide that there is no significantly different in MPG between cars with horsepower above 140 and those with 140 or below.

Q7

WHICH CAR MODEL YEAR HAS THE HIGHEST AVRERAGE FUEL EFFICIENCY (MPG)?





Report

mpg

model_year	Mean	N	Std. Deviation
70	14.400	1	.
72	30.000	2	.1414
73	30.225	4	7.2890
74	33.800	1	.
75	26.400	2	2.1213
76	29.075	4	7.1932
77	37.900	1	.
78	26.700	4	8.3491
80	36.700	2	.4243
81	19.250	4	10.1959
82	34.967	3	3.6199
Total	28.507	28	8.2180

THE ANALYSIS REPORTS AVERAGE FUEL EFFICIENCY FOR ALL MODEL YEARS. FROM THESE RESULTS, 1977 RECORDS THE HIGHEST MEAN MPG (37.7), WITH THE LINE AND BAR CHARTS ILLUSTRATING THE OVERALL TREND ACROSS YEARS.

Q8

DOES THE MODEL YEAR OF THE CAR SIGNIFICANTLY PREDICT THE MPG OF THE CAR?

1 SIMPLE LINEAR REGRESSION

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4.041	1	4.041	.058	.812[b]
	Residual	1819.417	26	69.978		
	Total	1823.459	27			

a. Dependent Variable: mpg

b. Predictors: (Constant), model_year

Assumptions	• Relationship between variables is linear • Residuals are normally distributed • Variance of residuals is constant (homoscedasticity)
H_0 **H1**	Model Year does not significantly predict MPG Model Year does significantly predict MPG
Test Name	ANOVA (Simple Linear Regression)
Output	Sig. .812[b]
Decision	Since the p-value = 0.812 > 0.05, we accept H_0.

Q9

WHICH FACTOR HAS THE MOST INFLUENCE ON MPG: WEIGHT, HORSEPOWER, OR CYLINDERS?

1 MULTIPLE LINEAR REGRESSION

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	254.491	3	84.830	1.245	.317[b]
	Residual	1567.696	23	68.161		
	Total	1822.187	26			

a. Dependent Variable: mpg

b. Predictors: (Constant), cylinder, weight, horsepower

Assumptions	• Linear relationship between predictors and MPG • Residuals are normally distributed • No multicollinearity • Homoscedasticity
H_0 **H1**	None of the variables weight, horsepower, or cylinders significantly influence MPG. At least one of the variables weight, horsepower, or cylinders significantly influences MPG.
Test Name	ANOVA (Multiple Linear Regression)
Output	Sig. .317[b]
Decision	Since p = 0.317 > 0.05, we accept H_0, meaning the model is not statistically significant.

Q9

1 MULTIPLE LINEAR REGRESSION

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance
1	(Constant)	35.298	11.288		3.127	.005	
	weight	-.005	.003	-.388	-1.762	.091	.773
	horsepower	.014	.059	.054	.233	.818	.699
	cylinder	.995	1.100	.203	.905	.375	.743

Assumptions	• Linear relationship between predictors and MPG • Residuals are normally distributed • No multicollinearity • Homoscedasticity
H_0 **H1**	None of the variables weight, horsepower, or cylinders significantly influence MPG. At least one of the variables weight, horsepower, or cylinders significantly influences MPG.
Test Name	Multiple Linear Regression
Output	-.388 .054 .203
Decision	Although the model is not statistically significant overall, weight has the strongest influence on MPG, because it has the largest absolute standardized Beta value.

STAT 231

THANK YOU

